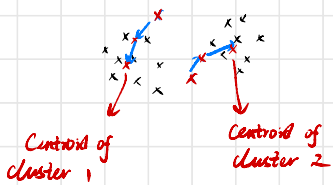
K-means is a simple and efficient clustering method

Formulation: Suppose we have data points $x_1, \dots, x_n$, where $x_i \in \mathbb{R}^d$. For each of the data points, we introduce $p_i \in \{1, \dots, k\}$ to indicate which cluster point i should be assigned.

Example:



Target:

$$\underset{\{p_1, \dots, p_n\}}{\operatorname{argmin}} \sum_{j=1}^{k} \sum_{i=1}^{n} \mathbb{I}(p_i = j) \|x_i - \mu_j\|$$

where

$$\mu_j = \frac{1}{\sum_{i=1}^{n} \mathbb{I}(p_i = j)} \sum_{i=1}^{n} \mathbb{I}(p_i = j)\, x_i$$

Algorithm:

1. Randomly initialize $\mu_1^{(0)}, \dots, \mu_k^{(0)}$

2. E-step: Compute the distance of each centroid to each cluster and let

$$p_i^{(k)} = \underset{j}{\operatorname{argmin}} \|x_i - \mu_j^{(k)}\|^2$$

3. M-step:

$$\mu_j^{(k+1)} = \frac{1}{\sum_{i=1}^{n} \mathbb{I}(p_i^{(k)} = j)} \sum_{i=1}^{n} \mathbb{I}(p_i^{(k)} = j)\, x_i$$

Notes about the algorithm:

1. The algorithm is guaranteed to converge since both E-step and M-step are guaranteed to lead to lower loss. (greedy algorithm)
2. The final results rely on initialization. (The algorithm might not converge to the true results)

In practice, we could initialize the centroid multipe times, run the algorithm, and pick the one with smallest lose.

However, this is not very computationally efficient.

A Better approach is to carefully initialize the centroids:

K-means ++

1. Choose randomly from $x_1, \dots, x_n$ as $u_1^{(0)}$

2. choose next centroid from $x_1, \dots, x_n$ with probability

$$\frac{D^2(x_i)}{\sum_{i=1}^{n} D^2(x_i)},$$

where $D(x_i)$ is the distance of x_i and the closest centroid that has been chosen.

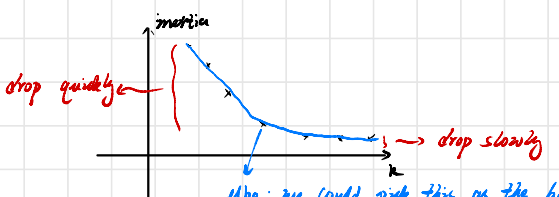
3. Repeat 2 until choose all k centroids

Evaluate k-means algorithm and choose k.

We obviously could use loss function to evaluate the results (inertia)

However it is not good for choosing k.

Increase k ↑ → lower inertia ↓ (the solution for k is a special case for k+1)



However, elbo is not always so obvious.

A better metric: silhouette coefficient

An instant silhouette score is computed as $\frac{b-a}{\max(b,a)}$ (the larger the better)

- b: the average distance to the points in the nearest cluster
- a: the average distance to the points in its own cluster.

Then we only need to choose k with largest average silhouette.

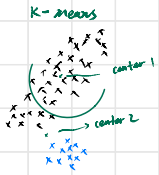
Pros of K-means: 1. Simple and efficient
2. Guaranteed convergence

Cons: 1. require manual choose of k

2. sensitive to initialization

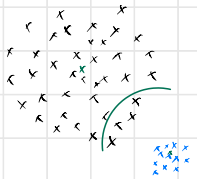
3. Assume spherical clusters and same variance

4. Sensitive to outliers.



K-means cannot handle elliptical clusters well

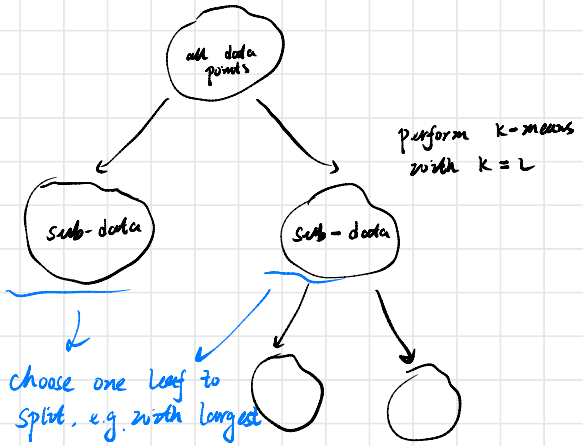
because it creats circles or balls around its centers



K-means cannot deal with clusters with different variance well

Because it uses the same distance measure (normally L_2) for different clusters

Hierarchical K-means



until reach given K / maximum tree-depth, minimum cluster size / variance threshold ...

Compared with Standard K-means, Hierarchical K-means have the following advantages:

1. Less sensitive to initialization of centroids.
   (every step it only need to solve K = 2, which is simpler)

2. It works better with unequal variance clusters.

